

Challenge 1 — Values-Led Problem Statement

A Domain-Specific Small Language Model for Type 2 Diabetes Self-Management Support

Question 1. Describe the problem that the selected project is trying to solve.

People living with type 2 diabetes have to make daily decisions about food, medication, and symptoms almost entirely on their own, in the long stretches of time between clinic visits. In practice, very few of them ever receive proper, structured education on how to manage the condition; only 6.8% of newly diagnosed patients get this within their first year, and across low and middle income countries, no diabetes self-management education programme currently meets International Diabetes Federation standards. Left to fill that gap, patients turn to whatever is on hand: family advice, guesswork, or the internet, including AI chatbots, which have been shown to give unsafe or problematic answers to real patient questions a meaningful share of the time. The result is a daily information and support gap for people who are already carrying the weight of a chronic condition.

Question 2. Who is affected by this problem? Who experiences the problem?

- Patients themselves, who make unsupervised daily choices about food, insulin or oral medication dosing, and worrying symptoms without a clinician on hand to ask.
- Family members, who research shows shape self-management in both directions and sometimes provide crucial support, but sometimes add control, conflict, or financial strain.
- Community health workers, particularly in settings like South African ward-based teams, who are often the ones actually delivering diet, medication, and self-care education and could use a reliable tool to support that work.

The burden falls hardest in low- and middle-income countries, which are home to 79% of people living with diabetes worldwide, and where diabetes incidence has risen by over 90% since 1990, even as formal education and support infrastructure has not kept pace.

Question 3. Why is solving this problem important? Describe the consequences if the problem is not solved.

Diabetes self-management is not a side detail; it is most of what actually determines a person's health outcomes, since clinicians only see patients for a few minutes every few months. When patients report the biggest obstacles to self-care, difficulty understanding medical terms (56.9%) and distrust of online sources (54.4%) rank just behind lack of time; both barriers that better information can genuinely address. Left unsolved, the consequences are measurable: studies report that a fifth to two-fifths of patients in LMIC settings show poor or inadequate self-care. Poor self-care compounds over years into avoidable complications, hospitalizations, and lower quality of life. And the fallback option many patients already reach for generic AI chatbots is not a safe substitute, since even leading models were judged to give unsafe answers to 5 to 13% of patient-posed questions, and problematic answers far more often than that.

Question 4. Which values should guide your solution? Identify the values that should guide your project.

- Scope humility: the system should know, and act on, the limits of what it is safe to answer.
- Dignity, not paternalism: patients should receive real, cited information rather than a watered-down or withheld answer.

- Communal accountability: answers should be built to be shared with and discussed by a caregiver or health worker, not consumed in isolation.
- Safety and trustworthiness: every answer should be grounded in recognized clinical guidance (ADA, WHO/IDF) with traceable sources, and any signs of a medical emergency should be escalated rather than answered.

Question 5. Why did you choose these values?

These values follow directly from the evidence in our problem research, rather than being chosen in the abstract. Scope humility responds directly to the finding that even the best existing chatbots still produced unsafe responses to a meaningful share of real patient questions; the risk is real and documented, not hypothetical. Dignity, not paternalism, responds to the fact that patients' top reported obstacles are comprehension and trust, not a need to be shielded from information; an overly cautious system that refuses to engage would fail patients differently. Communal accountability reflects the systematic-review finding that families and community health workers already shape diabetes self-management in practice, so a solution that ignores that reality, or assumes involvement is always welcome, would misread the environment it is entering.

Question 6. What ethical risks should your team consider? Every AI system has risks.

- Over-reliance: patients may come to trust the system beyond its actual scope, especially given documented rates of unsafe chatbot responses in this domain.
- Family involvement cutting both ways: shareable output is supportive in some households but could be used to control or burden a patient in others; the system cannot assume involvement is always welcome.
- The concise-versus-caveated tension: an answer safe enough to include every possible caveat often becomes too long to actually be read and used; this tradeoff is real and not fully resolved.
- Missing the biggest barrier: the most-reported obstacle patients describe is simply lack of time, and an AI assistant does not create more of it; the project only addresses comprehension and trust, not time.

Question 7. How will your team reduce these risks?

- Hard-coded escalation layer: queries describing possible severe hypoglycaemia, hyperglycaemia, or DKA are rerouted straight to a "seek care now" message before any answer is generated, reducing the chance of over-reliance in genuine emergencies.
- Grounded, cited answers: responses are retrieved from ADA and WHO/IDF guidance with traceable sources rather than generated as a confident, unsupported guess.
- Shareable, explainable framing: output is phrased to be carried to and discussed with a caregiver or community health worker, encouraging conversation rather than assuming private, unmediated use while we remain explicit in messaging that family involvement should not be assumed or forced.
- Narrow scope, stated honestly: we explicitly name what the system does not do, and it does not save patients time, and it does not replace clinical care, so users' expectations are calibrated rather than inflated.

Question 8. What outcomes would define success? Success is more than model accuracy.

- Patients and community health workers report better understanding of guideline information after using the tool, addressing the 56.9% comprehension barrier directly.
- Patients report increased trust in the information they receive, addressing the 54.4% distrust-of-online-sources barrier.

- The escalation layer reliably routes emergency pattern queries to “seek care now” with a very low false-negative rate, since even one missed severe case is a serious failure.
- Answers are actually shared with and discussed alongside caregivers or health workers, rather than consumed in isolation, matching the communal-accountability goal.
- No evidence that the tool increases family conflict or over-reliance beyond its stated scope, tracked as an explicit risk metric, not just a hoped-for side effect.
- The system is honest about its own limits in its answers, rather than optimizing purely for user engagement or answer length.

Question 9. Final values-led problem statement (100–150 words).

Across low- and middle-income countries, most people living with type 2 diabetes make daily food, medication, and symptom decisions alone between clinic visits, with only 6.8% receiving structured self-management education and no LMIC program meeting International Diabetes Federation criteria. This gap affects patients, the families who share the burden of care, and the community health workers who support them. Left unaddressed, it drives inconsistent self-care and preventable complications. Our project builds a narrow-scope, guideline-grounded small language model that answers diabetes self-management questions in plain language, cites its sources, and escalates any sign of medical emergency to “seek care now” rather than generating advice. Guided by scope humility, dignity rather than paternalism, and communal accountability, the system produces shareable, explainable answers meant to be discussed with a caregiver or health worker, building comprehension and trust, not replacing clinical care or the human relationships around it.

Question 10. Names of Active Team Members.

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